PADHAKU
Curiosity · Class 6 Science



CHAPTER 2

Diversity in the Living World

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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⭐ Diversity in the Living World (introduction)

? **Have you ever observed different plants and animals around you? Share and discuss your observations with your friends and teacher.**

💡 SAMPLE ANSWER

Yes. Around my home and school I see grass, tulsi, neem and mango trees.

I see crows, sparrows, ants, butterflies, cows and dogs.

Each plant and animal looks different, and each bird has its own chirp. This is **diversity** in nature.

⭐ 2.1 Diversity in Plants and Animals Around Us

Activity 2.1 **Activity 2.1: Let us explore and record**

Complete Table 2.1 (observations of different plants) for hibiscus, neem and any other plant.

💡 SAMPLE ANSWER

Plant	Stem	Leaves	Flowers	Other features
Common grass	Soft and thin	Single leaf grows alternately from the stem	—	Green leaves
Tulsi	Hard and thin	Pairs of leaves in opposite directions	Pinkish purple	Leaves have a strong smell
Hibiscus	Hard	Broad leaves with jagged edges, net-like veins	Large, red	Shrub
Neem	Hard and thick	Many small leaflets with toothed edges	Small, white	Leaves are bitter; big tree

Your own observations may be different (sample).

? **What similarities and differences did you find among the plants that you observed?**

✔ ANSWER

Similarities: all plants have roots, a stem, green leaves; most have flowers.

Differences: stems are tall or short, hard or soft; leaves have different shapes and arrangement on the stem; flowers differ in colour, shape and smell.

Table 2.2 **Complete Table 2.2: place where the animal lives, food it eats, the way it moves, other features (crow, ant, cow, any other).**

💡 SAMPLE ANSWER

Animal	Lives in	Food	Movement	Other features
Crow	Tree	Insects	Fly and walk	Carries a twig in its beak
Ant	Nest in soil and burrow	Leaves, seeds and insects	Walk (crawl)	Has six legs
Cow	Shed, fields	Grasses, leaves	Walk	Has horns and a tail
Fish	Water (pond, river)	Small plants and insects	Swim	Has fins
Squirrel	Tree	Nuts, fruits, seeds	Run and climb	Bushy tail

? **What are the similarities and differences among the animals that you have observed and recorded in Table 2.2?**

✔ ANSWER

Similarities: all need food, water and a place to live; all can move.

Differences: some live on land, some on trees, some in water (fish), some both on land and in water (frog). They eat different foods (grass, insects, seeds) and move in different ways (fly, walk, crawl, swim).

? **Sketch the plants and animals you observed or make a scrapbook. While travelling to and from school, add any new plant or animal to Tables 2.1 and 2.2.**

💡 SAMPLE ANSWER

Do it in your notebook. Example new entries: **peepal** (tall tree, heart-shaped leaves), **marigold** (herb, orange flowers), **sparrow** (lives in trees/houses, eats seeds, flies), **earthworm** (lives in soil, crawls).

Activity 2.2 (a) **Activity 2.2: Let us appreciate**

What are your observations about the various plants and animals that have been drawn?

💡 SAMPLE ANSWER

The drawings show many different plants (trees, herbs, flowering plants) and animals (birds, insects, pets, wild animals).

Each one has a different shape, size and colour. This shows the variety (diversity) of living things.

Activity 2.2 (b) **How many different plants and animals did the entire class draw on the blackboard?**

💡 SAMPLE ANSWER

Count the drawings in your class. For example, a class of 30 may draw about 15 different plants and 20 different animals (some are repeated).

Activity 2.2 (c) **Do you think that there may be many more varieties of plants and animals other than those drawn on the board?**

✔ ANSWER

Yes. There are many more plants and animals in forests, deserts, mountains and oceans that we have never seen.

The variety of plants and animals in a region is its **biodiversity**.

? **Trees give food and shelter to birds; animals spread seeds after eating fruits. Can you think of more such examples?**

✔ ANSWER

- Bees and butterflies take nectar from flowers and help in pollination.

- Cow dung and dead leaves make the soil fertile for plants.
- Plants give oxygen and food to animals.
- Birds build nests on trees and eat insects that harm plants.

So plants and animals **depend on each other**.

⭐ 2.2 How to Group Plants and Animals?

? How would you arrange your books and notebooks in groups? Would arranging them in groups help you better organise your school bag?

✔ ANSWER

I can group them by subject (all Science books and notebooks together), or by size, or by the day's timetable.

Yes. Grouping helps me find things quickly and keep the bag tidy.

Activity 2.3 **Activity 2.3: Let us group**

Group the plants and animals on the cards on the basis of common features. Share the basis of grouping with other groups.

💡 SAMPLE ANSWER

Plants: with flowers / without flowers; hard stem / soft stem; tall / short.

Animals: by food (plant-eaters, flesh-eaters); by place they live (land, water, trees); by movement (fly, walk, swim, crawl); by colour.

? The basis used by different groups may vary. What do you think are the reasons behind it?

✔ ANSWER

Different students chose **different common features** for grouping. For example, some chose the height of plants, others chose presence or absence of flowers.

? What is the importance of grouping?

✔ ANSWER

Grouping makes it **easier to understand and study** plants and animals on the basis of their similarities and differences.

⭐ 2.2.1 How to group plants?

Activity 2.4

Activity 2.4: Let us group

Fill Table 2.3: height, nature of stem, where branches arise and name of plant group.

💡 SAMPLE ANSWER

Plant	Height	Stem colour	Tender/Hard	Thick/Thin	Branches from	Group
Mango	Tall	Brown	Hard	Thick	Higher up on the stem	Tree
Rose	Medium	Brown	Hard	Thin	Close to the ground	Shrub
Tomato	Short	Green	Tender	Thin	Close to the ground	Herb
Neem	Tall	Brown	Hard	Thick	Higher up on the stem	Tree
Hibiscus	Medium	Brown	Hard	Thin	Close to the ground	Shrub
Mint (pudina)	Short	Green	Tender	Thin	Close to the ground	Herb

? What differences do you observe among herbs, shrubs, and trees? How can you group plants as herbs, shrubs, and trees based on the data in Table 2.3?

✔ ANSWER

Feature	Herb	Shrub	Tree

Height	Short	Medium	Tall
Stem	Green, soft (tender), thin	Brown, hard, woody, not very thick	Brown, hard, woody, thick
Branches	Close to the ground	Many stems branching close to the ground	Start higher up on the stem
Example	Tomato	Rose	Mango

? **What can be other features on the basis of which you can group plants?**

✔ ANSWER

- Type of stem: weak stems that need support (**climbers**) or that spread on the ground (**creepers**).
- Leaves: shape and **venation** (pattern of veins).
- Roots: **taproot** or **fibrous roots**.
- Seeds: one or two **cotyledons**; presence or absence of flowers.

Activity 2.5 **Activity 2.5: Let us compare**

Do you notice any variation in the shape and structure of the leaves you collected?

✔ ANSWER

Yes. Leaves differ in shape (broad, long and narrow, heart-shaped), size, colour, edges (smooth or toothed) and in the pattern of veins on them.

? **What differences do you see in the veins of leaves shown in Fig. 2.4a (hibiscus) and Fig. 2.4b (banana)?**

✔ ANSWER

Hibiscus leaf: a net-like pattern of veins on both sides of a thick middle vein: **reticulate venation**.

Banana leaf: veins run parallel to each other: **parallel venation**.

? **Do you think that plants can be grouped on the basis of venation present in their leaves?**

✔ ANSWER

Yes. Plants with reticulate venation (hibiscus, mango) form one group and plants with parallel venation (banana, grass) form another.

? **Do all plants have roots? Are these roots similar?**

✔ ANSWER

Almost all plants we see around us have roots. **No**, the roots are not similar: some plants have a **taproot** and others have **fibrous roots**.

Activity 2.6 (a) **Activity 2.6: Let us find out**

What are the similarities and differences in the roots of the plants collected by you?

✔ ANSWER

Similarity: all roots grow below the ground, hold the plant and are not green.

Difference: some plants have one thick main root with small side roots; others have a bunch of thin roots of similar size.

Activity 2.6 (b) **What differences do you see in the roots of plants shown in Fig. 2.5a (mustard) and Fig. 2.5b (grass)?**

✔ ANSWER

Mustard: one main root with small side roots arising from it: **taproot**.

Grass: a bunch of similar-sized thin roots arising from the base of the stem: **fibrous roots**.

? **Does your collection include any other grasses? What kind of roots do they have?**

✔ ANSWER

Yes (e.g. doob grass, wheat seedling). All grasses have **fibrous roots**.

? **Is there any relation between the type of leaf venation and the type of root of the same plant? How do we find this out?**

✔ ANSWER

Yes. We can find out by observing the leaves and roots of many plants and recording them in a table (Activity 2.7).

Plants with **reticulate venation** generally have **taproots**; plants with **parallel venation** generally have **fibrous roots**.

Activity 2.7 (a) **Activity 2.7: Let us relate and analyse**

Record leaf venation and type of root of five saplings in Table 2.4.

💡 SAMPLE ANSWER

Plant	Leaf venation	Type of root
Lemongrass	Parallel	Fibrous
Marigold	Reticulate	Taproot
Sadabahar (periwinkle)	Reticulate	Taproot
Tulsi	Reticulate	Taproot
Maize	Parallel	Fibrous

Activity 2.7 (b) **Do you observe any relation between the leaf venation and types of root in these plants? Do other plants with reticulate venation have taproots too? Do other plants with parallel venation have fibrous roots too?**

✔ ANSWER

Yes. Generally, plants with **reticulate venation have taproots** (e.g. sadabahar, chickpea), and plants with **parallel venation have fibrous roots** (e.g. lemongrass, wheat).

? **Is there any relation among the seed of a plant, types of root and leaf venation? Are all seeds similar?**

✔ ANSWER

No, all seeds are not similar. Some seeds have two cotyledons (chickpea), some have one (maize).

Yes, there is a relation (see Activity 2.8).

Activity 2.8 (a) **Activity 2.8: Let us compare**

Observe the structure of soaked chickpea and maize seeds. Are they similar or different?

✔ ANSWER

They are **different**.

The chickpea seed splits into **two parts** (two cotyledons): it is a **dicot** seed.

The maize seed has a **single thin cotyledon**: it is a **monocot** seed.

Activity 2.8 (b) **What relation do you observe among leaf venation, root types and the number of cotyledons in seeds of a plant?**

✔ ANSWER

Plant type	Cotyledons	Leaf venation	Root
Dicot (e.g. chickpea)	Two	Reticulate	Taproot
Monocot (e.g. maize, wheat)	One	Parallel	Fibrous

⭐ 2.2.2 How to group animals?

? **How can we group such a wide variety of animals? What features can you think of to group them?**

✔ ANSWER

We can group animals by: the food they eat, the place where they live, the way they move, the body parts used for movement, their shape, size, structure and colour.

Activity 2.9 (a) **Activity 2.9: Let us find out**

Which body parts are used by the animals in your poster and in Fig. 2.7 for movement?

💡 SAMPLE ANSWER

- Legs: goat, cow, dog, ant (walk, run, jump)

- Wings and legs: birds, housefly, butterfly (fly and walk)
- Fins and tail: fish (swim)
- Whole body (muscles): snake, earthworm (crawl)

(check the picture in your book)

Table 2.5 **Complete Table 2.5: type of movement and body parts used for movement.**

💡 SAMPLE ANSWER

Animal	Type of movement	Body parts used
Ant	Walks (crawls)	Legs
Goat	Walks and jumps	Legs
Pigeon	Walks and flies	Legs and wings
Housefly	Walks and flies	Legs and wings
Fish	Swims	Fins
Snake	Crawls	Whole body
Frog	Hops/jumps and swims	Legs
Horse	Walks and runs	Legs

? **What conclusions can you draw from the data given in Table 2.5?**

✔ ANSWER

Different animals have **different types of movement**: they fly, run, crawl, walk, hop, jump or swim.

They use **different body parts** for moving: wings, legs, fins and others.

? **How can we group animals based on their movements?**

✔ ANSWER

- Animals that fly: pigeon, crow, housefly
- Animals that walk/run: goat, cow, horse
- Animals that swim: fish
- Animals that crawl: snake, earthworm
- Animals that hop/jump: frog

⭐ 2.3 Plants and Animals in Different Surroundings

? Does the movement of these animals depend upon their surroundings? (Example: fish and goat)

✔ ANSWER

Yes. Fish live in water; they have a streamlined body and fins to swim.

Goats live in grassy areas on land; they walk with the help of legs.

Activity 2.10 **Activity 2.10: Let us compare and analyse**

List plants and animals found in the desert, on mountains, in the ocean, in the forest and any other region (Table 2.6).

💡 SAMPLE ANSWER

Desert	Mountains	Ocean	Forest	Other (pond)
Camel	Deodar tree	Fish	Lion	Frog
Cactus	Yak	Whale	Tiger	Lotus
Desert fox	Rhododendron	Octopus	Elephant	Duck

? What are your observations regarding plants and animals found in various regions?

✔ ANSWER

The plants and animals found in one kind of region are **different** from those found in another kind of region. Each has features that help it live in its own region.

? Cactus (desert) and deodar (mountains) are different from each other. Why is it so? Why does the biodiversity of a region vary from that of another?

✔ ANSWER

Conditions are different in each region.

- Desert: very little water, very hot days and cold nights. Cactus has **thick, fleshy stems that store water**.

- Cold mountains: frequent snowfall. Deodar is **conical with sloping branches**, so snow slides off easily.

So biodiversity varies from region to region because of **diverse conditions**.

? **Look at a camel from the hot desert of Rajasthan and a camel from the cold desert of Ladakh. What differences do you observe? What advantages do these differences provide?**

✔ ANSWER

Hot desert camel	Cold desert camel
Long legs with wide hooves	Shorter height and legs
One hump	Two humps
Short hair	Long hair from head to neck

Advantages: long legs and wide hooves help walk on sand without sinking; short legs help walk easily in mountains; humps store food for times of scarcity; long hair helps survive the cold winters of Ladakh.

? **What other features can help camels to survive in the desert?**

✔ ANSWER

Camels excrete **small amounts of urine**, their **dung is dry** and they **do not sweat**.

So they do not lose much water and can survive for many days without drinking water.

? **What are the different ways in which you can group plants and animals based on their habitats?**

✔ ANSWER

- **Terrestrial** (live on land): forests, deserts, grasslands, mountains, e.g. camel, goat, deodar.
- **Aquatic** (live in water): ponds, lakes, rivers, oceans, e.g. fish, whale, lotus.
- Live both in water and on land (**amphibians**), e.g. frog.

? **What would happen if the habitat of a plant or an animal is damaged? What would happen if a goat does not get grass to eat? Can a fish survive without water?**

✔ ANSWER

If a habitat is damaged, plants and animals **lose their homes, food and other resources**; many may die or move away. This leads to **loss of biodiversity**.

A goat without grass will not get food and will become weak and may die.

No, a fish cannot survive without water.

? **Check with your parents, grandparents and neighbours about plants, birds, insects or other animals they used to see often in their childhood but do not see as often now.**

💡 SAMPLE ANSWER

Example: grandparents say they often saw **sparrows, vultures, fireflies, frogs** and some wild fruit trees, which are rare now.

Reason: their habitats are damaged by cutting trees, buildings, pollution and pesticides.

? **More to know: Find out about the sacred groves in your region.**

💡 SAMPLE ANSWER

Ask elders or your teacher. Sacred groves are forest patches protected by the local community; no one cuts trees or harms animals there.

Examples in India: **Mawphlang** (Meghalaya), **kavus** (Kerala), **devrai** (Maharashtra), **orans** (Rajasthan).

🔭 Let us enhance our learning

1 **Here are two types of seeds: (a) Wheat (b) Kidney beans. What differences do you find among the roots and leaf venation of their plants?**

✔ ANSWER

Plant	Seed	Root	Leaf venation
(a) Wheat	Monocot (one cotyledon)	Fibrous roots	Parallel
(b) Kidney beans (rajma)	Dicot (two cotyledons)	Taproot	Reticulate

2 **Group the animals by habitat: aquatic in 'A', terrestrial in 'B', animals living in both habitats in 'C'.**

Horse, Dolphin, Frog, Sheep, Crocodile, Squirrel, Whale, Earthworm, Pigeon, Tortoise

✔ ANSWER

A (aquatic): Dolphin, Whale

B (terrestrial): Horse, Sheep, Squirrel, Earthworm, Pigeon

C (both): Frog, Crocodile, Tortoise

(Tortoise is put in C as the book's answer; a true land tortoise lives only on land, while turtles live in water and come on land.)

3 **Manu's mother told him that radish is a kind of root. Examine a radish and write what type of root it is. What type of venation would you observe in the leaves of radish plant?**

✔ ANSWER

Radish is a **taproot**: one thick main root (swollen with stored food) with thin side roots.

So its leaves have **reticulate venation**.

4 **Look at the image of a mountain goat and a goat found in the plains. Point out the similarities and differences between them. What are the reasons for these differences?**

💡 SAMPLE ANSWER

Similarities: both have four legs with hooves, horns, a similar body shape, and both eat grass and leaves.

Differences: the mountain goat has **thick, long hair (fur)**, strong legs and hooves and bigger horns; the plains goat has short, thin hair.

Reasons: mountains are very cold, so thick hair keeps the goat warm; strong hooves help it climb and grip steep rocky slopes. These are **adaptations** to its habitat.

(check the picture in your book)

5 **Group the following animals into two groups based on any feature other than those discussed in the chapter: cow, cockroach, pigeon, bat, tortoise, whale, fish, grasshopper, lizard.**

💡 SAMPLE ANSWER

Feature: **presence of a backbone**.

- With backbone: cow, pigeon, bat, tortoise, whale, fish, lizard
- Without backbone: cockroach, grasshopper

(Another way: animals that lay eggs: cockroach, pigeon, tortoise, fish, grasshopper, lizard; animals that give birth to young ones: cow, bat, whale.)

6 **As the population grows and people want more comfortable lives, forests are being cut down. How can this affect our surroundings? How do you think we can address this challenge?**

💡 SAMPLE ANSWER

Effects:

- Plants and animals lose their habitat, food and shelter; biodiversity is lost.
- Less rain, more heat, soil erosion and floods; air becomes polluted.
- Wild animals may come into villages and towns.

What we can do:

- Plant more trees and stop unnecessary cutting of trees.
- Save and recycle paper and wood.
- Protect forests, sacred groves and protected areas; spread awareness.

7 **Analyse the flowchart. What can be examples of 'A' and 'B'? Plant → Does it have leaves? → Yes → Does it have reticulate venation? → Yes → A; No → B**

💡 SAMPLE ANSWER

A (leaves with reticulate venation, dicots): hibiscus, mango, mustard, chickpea, rose.

B (leaves with parallel venation, monocots): wheat, maize, grass, banana, lemongrass.

(check the picture in your book)

8 **Raj argues with his friend Sanjay that "Gudhal (hibiscus) plant is a shrub." What questions can Sanjay ask for clarification?**

💡 SAMPLE ANSWER

- Is the plant shorter than a tree (medium height)?
- Is its stem brown, hard and woody?
- Is the stem thinner than the trunk of a tree?
- Do its branches start close to the ground?

If the answers are yes, Raj is right: hibiscus is a **shrub**.

9 **Find examples of these plants for each group:**

Group	Type of seed	Type of root	Examples
A	Dicot	Taproot	?
B	Monocot	Fibrous roots	?

✔ ANSWER

A: chickpea, mustard, kidney bean, hibiscus, mango

B: wheat, maize, rice, grass, lemongrass

9 (a) **What other similarities do plants of group A have?**

✔ ANSWER

Their leaves have **reticulate venation** and their seeds have **two cotyledons**.

9 (b) **What other similarities do plants of group B have?**

✔ ANSWER

Their leaves have **parallel venation** and their seeds have **one cotyledon**.

10 **Observe the labelled part (webbed feet) of a duck. What differences do you observe in the feet of the duck compared to the other birds (pigeon)? Which activity would the duck be able to perform using this part?**

✔ ANSWER

The duck's toes are **joined by a thin skin (webbed feet)**; the pigeon's toes are separate, for walking and holding branches. Webbed feet work like paddles, so the duck can **swim** in water easily.

📘 Learning further

1 **Read about one Indian scientist or a wildlife biologist who is working towards protection of India's biodiversity. Prepare a brief report.**

💡 SAMPLE ANSWER

Sample: **Salim Ali**, the 'Birdman of India', studied birds across India, wrote 10 books on birds and helped protect Keoladeo National Park (Bharatpur). Write his work, places and why it matters in 5–6 lines.

2 **Explore the contributions of Divya Mudappa, Usha Lachungpa, Ghazala Shahabuddin, Nandini Velho, Vidya Athreya, Uma Ramakrishnan and Divya Karnad towards biodiversity in India. Prepare a report on any three of them.**

💡 SAMPLE ANSWER

Search in the library or internet with your teacher. For each of the three, write: name, field of work (e.g. rainforests, wildlife, leopards, tigers, oceans/fisheries), where she works, and how her work helps protect biodiversity.

3 **Label the plants in your school with their local names with the help of your teacher or the gardener. List them in your notebook.**

💡 **SAMPLE ANSWER**

Make small name cards (e.g. Neem, Peepal, Tulsi, Gudhal, Gulmohar) and list them in your notebook with the type of plant (tree, shrub, herb).

4 **Plan a field visit or a nature walk with your teacher. Record your observations. Prepare a class biodiversity register.**

💡 **SAMPLE ANSWER**

In the register, make columns: date, place, name of plant/animal, type (tree/shrub/herb, bird/insect…), where seen, special features, observed by.

5 **Find out about 'Project Tiger' and other similar projects initiated in India to protect our biodiversity. Prepare a presentation for your class.**

💡 **SAMPLE ANSWER**

- **Project Tiger** (1973): to protect the declining population of the Bengal Tiger.
- **Cheetah Reintroduction Project** (2022): to restore the population of the Cheetah.
- Habitats of the **Great Indian Bustard** declared Protected areas in Gujarat, Rajasthan and Maharashtra.

Add pictures and one slide on how students can help.

6 **In groups of six, discuss how you can protect biodiversity around you. Prepare a group-wise report of suggestions.**

💡 **SAMPLE ANSWER**

Sample suggestions: plant and care for trees; do not pluck flowers or disturb birds' nests; keep water for birds in summer; do not litter or use plastic; save paper; make a school garden; protect local ponds and sacred groves.

7 **Interact with elders to find out plants and animals they see now but were not seen earlier, and vice versa. Make a scrapbook with pictures.**

💡 **SAMPLE ANSWER**

Seen earlier, rare now (example): sparrows, vultures, fireflies.

Seen now, not earlier (example): new garden plants such as Parthenium (congress grass) and Lantana weeds, more pigeons in cities.

Paste pictures and write 2 lines about each.

🎉 **Chapter 2 done! Revise the answers once more tomorrow.**